Lab 7: Coral Reef Mapping

Kendall Fitzgerald

March 23, 2025

Introduction & Methodology

This lab explores how to implement the Allen Coral Atlas (ACA) and Google Earth Engine (GEE) data into ArcGIS to classify benthic habitats and geomorphic zones of coral reefs in two Marine Protected Areas (MPAs), one in Monte Cristi National Park in the Dominican Republic and one in Cahuita National Park in Costa Rica. We first uploaded the provided ACA GeoJSON files to QGIS, then converted them to a GeoPackage that was readable within ArcPro. As an alternative exploratory method, we then used GEE to edit code to export ACA data to raster files that could be used within ArcGIS.

Map 1 shows the results of exporting both the ACA benthic habitat raster and the thirteen categorized geomorphic zones of the coral reef within Monte Cristi NP and Table 1 describes the most widespread combinations within this area. Map 2 shows the benthic habitat from downloaded ACA data for Monte Cristi NP and Table 2 describes the proportion of each class (in square kilometers) that are located within the boundaries. Benthic habitat refers to the type of bottom that the seafloor can be classified as. In this case, the ACA uses “Rock”, “Coral/Algae”, “Rock”, “Seagrass”, and “Coral/Algae as descriptors. Geomorphic zone refers to ACA’s definition of a distinct area or habitat defined by its physical characteristics like depth, slope, and exposure to waves.

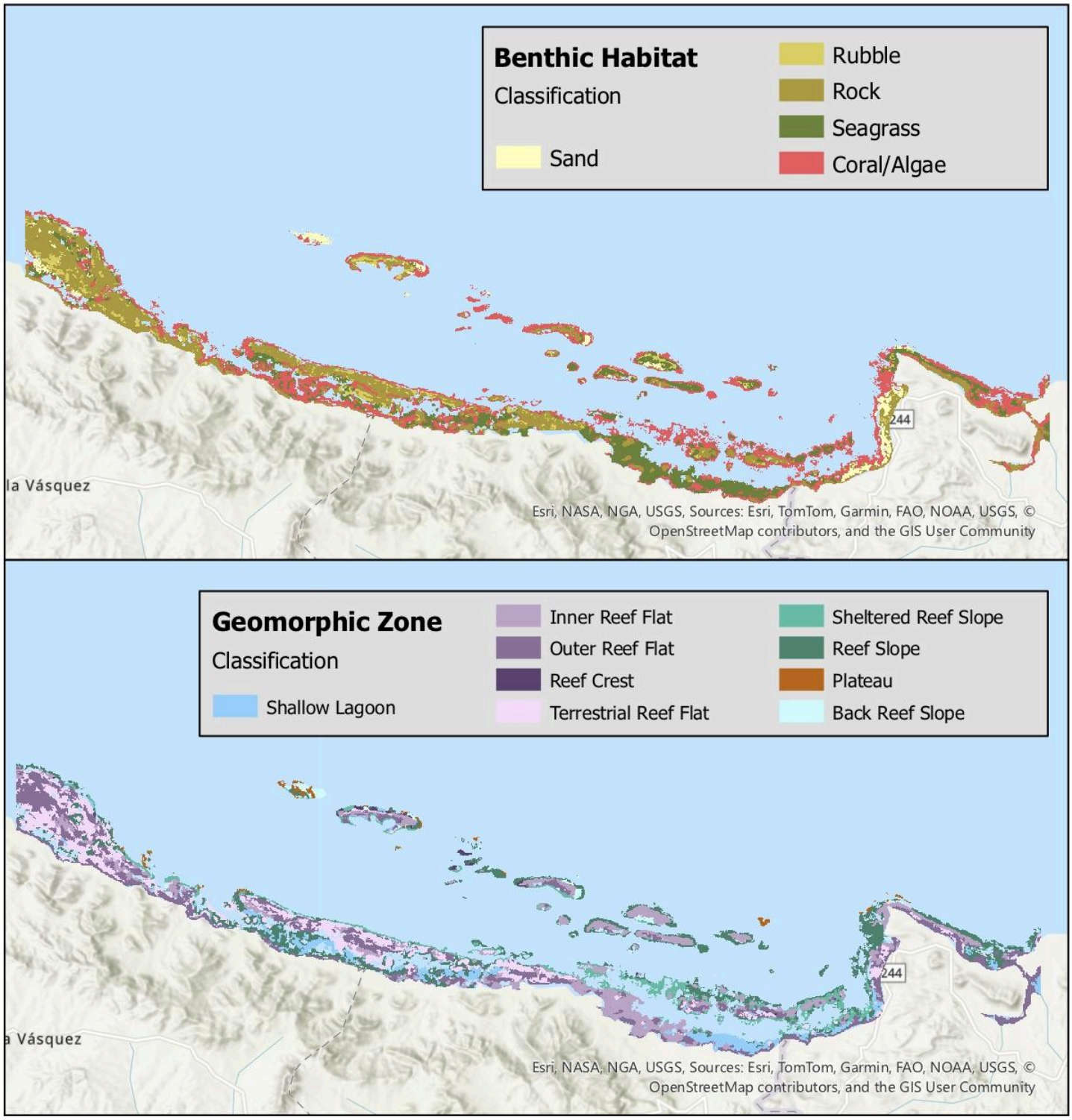
Table 1. *Table showing the 5 most widespread combinations of benthic habitat and geomorphic zone within Cristi National Park in the Dominican Republic.*

5 most widespread combinations in Monte Cristi National Park, DR			
Rank	***Benthic Habitat***	***Geomorphic Zone***	***Area (square km)***
1	Rock	Outer Reef Flat	7.874
2	Coral/Algae	Reef Slope	6.879
3	Rock	Terrestrial Reef Flat	4.431
4	Seagrass	Inner Reef Flat	3.882
5	Coral/Algae	Sheltered Reef Slope	2.919

Table 2. *Table showing the proportion of benthic habitat types of the coral reef within the boundaries of Cahuita National Park in Costa Rica.*

Cahuita National Park Benthic Habitat	
Type	***Area within Boundaries (square km)***
Sand	0.452
Rubble	0.305
Rock	0.840
Seagrass	0.094
Coral/Algae	2.134

Benthic Habitat and Geomorphic Zones in Monte Cristi National Park, Dominican Republic



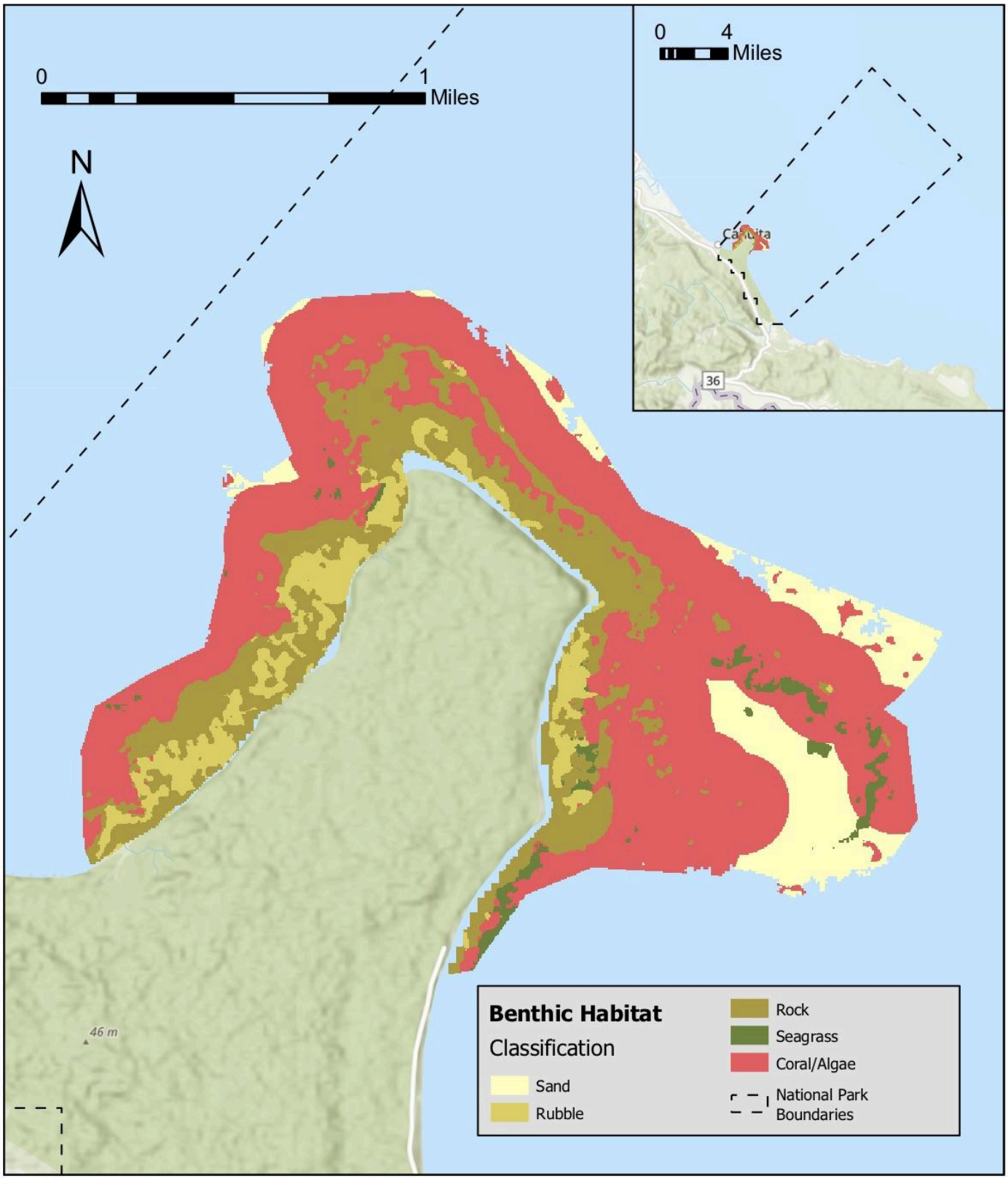
Map by Kendall Fitzgerald | Projection: WGS 1984 | March 24, 2025

0 5 Miles

N

Map 1. *This map shows benthic habitat classification and geomorphic zones of the coral reef within Monte Cristi National Park in the Dominican Republic.*

Benthic Habitat Classification of Coral Reef in Cahuita National Park, Costa Rica



Map by Kendall Fitzgerald Projection: WGS 1984 March 23, 2024

Esri, NASA, NGA, USGS, Esri, CGIAR, USGS, Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, © OpenStreetMap contributors, and the GIS User Community

Map 2. *This map shows benthic habitat classification of the coral reef within the Cahuita National Park boundaries in Costa Rica.*